

Ex Define Unitary matrices and show that if U is unitary then $\|Uz\| = \|z\|$ for any $z \in \mathbb{C}^n$.

Ex Define Hermitian matrices.

Ex If A is Hermitian, show that its eigenvalues are real and the eigenvectors of distinct eigenvalues are perpendicular.

Ex Let $A = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$ What is A^5 ? What are the eigenvalues & eigenvectors of A ?

Ex Diagonalize $\begin{bmatrix} 0 & 1-i \\ i+1 & 1 \end{bmatrix}$ & $\begin{bmatrix} 0 & -1+i \\ 1+i & i \end{bmatrix}$

Ex Show that $A^H A$ is Hermitian & positive def.

Ex Let $A = \begin{pmatrix} 0 & i & 0 \\ 0 & 0 & i \\ i & 0 & 0 \end{pmatrix}$ What is $A, A^2, A^3, A^4, \dots$?

What are the eigenvalues & eigenvectors.

Ex Solve $z^5 = 1-i$.